FairGate

Zero-knowledge bias-audit attestations for automated hiring tools

Prove a hiring algorithm passed the four-fifths rule.
Publish the verdict. Keep the people private.

Built on Midnight with Compact · Apache-2.0 · github.com/rifatsarwarcodes/Midnight_Buildathon

The problem: audits that cannot be both verifiable and private

- **NYC Local Law 144** and the **EU AI Act** require bias audits of automated hiring tools.
- The standard test is the EEOC **four-fifths rule**: every group's selection rate must be ≥ 80% of the best group's rate.
- Today the audit is a PDF.
  - Publish the counts → small groups become identifiable, hiring volumes leak to competitors.
  - Keep the counts → candidates, journalists and partners cannot verify anything.
- Nobody can check that the auditor's arithmetic was honest.

FairGate on Midnight

Private (witness data)

- applicants and selections per group
- which groups were too small to include
- the reference (best) group
- the commitment nonce
- the dataset and the model artifact

Public (ledger)

- tool id = H(employer, model hash)
- period, auditor id, threshold
- **exact floor of the minimum impact ratio**
- number of groups included
- sha-256 of the dataset, commitment to the counts

A failing tool cannot obtain a proof at all. A regulator can ask the auditor to open the commitment and verify the counts against chain state.

What the `attestAudit` circuit proves

For private counts `(a_i, s_i)`, inclusion mask, reference `(refA, refS)`, claimed floor `p`:

Check	Constraint (integer, no division)
shape	`s_i ≤ a_i`; `included_i ⇔ a_i ≥ minGroupSize`
coverage	`Σ a_i ≥ minTotal`, ≥ 2 groups included
reference is real and best	`(refA, refS)` ∈ included groups; `refS·a_i ≥ s_i·refA` ∀ i
four-fifths	`100·s_i·refA ≥ T·refS·a_i` ∀ included i, `T` read from the registry policy
exact floor	`100·s_i·refA ≥ p·refS·a_i` ∀ i and `100·s_j·refA < (p+1)·refS·a_j` for some j
binding	`countsCommit = persistentCommit([a, s, included], nonce)`

Seven circuits, six exported pure helpers so TypeScript derives identical ids and commitments.

Lying auditors, rejected by the circuit

Twelve hand-crafted workbooks in the test-suite, each refused by an assertion:

- exclude a large group that would fail → `a group large enough to include was excluded`
- fabricate a flattering reference group → `reference group is not one of the included groups`
- pick a weak reference so ratios look high → `does not have the highest selection rate`
- overstate or understate the published floor → `claimed minimum impact ratio is too high / too low`
- `selected > applicants`, too few applicants, one group, reference selected nobody
- attest with a weaker threshold → threshold is ledger state, not a prover input
- attest twice in a period, even via another auditor → period nullifier `H(toolId, period)`

40 tests total · `npm test` · CI compiles with Compact 0.31.1 on every push

Live on Midnight

- **Local devnet**: deploy → accredit → register → attest → verify in under two minutes; attestation proof lands in ~33 s.
- **Preprod**: registry deployed, see `DEPLOYMENTS.md` for addresses and transaction hashes.
- **CLI**: `admin accredit`, `employer register`, `auditor attest`, `verify tool | attestation | disclosure`.
- **Browser**: Verify (no wallet), Auditor workbench with private/public split, Employer, Regulator packet verifier, Admin. Lace wallet via DApp Connector 4.x; reads the ledger straight from the indexer.
- A tampered disclosure packet fails verification; an honest one reproduces both the commitment and the attested floor.

Who uses it

Party	What they do	What they learn
Employer	registers the exact model it runs	nothing new
Accredited auditor	proves the rule over private counts	nothing new
Candidate, journalist	checks a tool id on the Verify page	passed / revoked / none, floor ratio
Regulator	asks the auditor for the packet, verifies it on chain	the counts, and that they are the proven ones

Adoption path: start with LL144-covered employers in NYC (annual audits are already mandatory), sell to the auditors who issue them. Attestations become a portable, checkable credential for any AI hiring vendor.

Wave 3 and beyond

- **Intersectional audits**: 16-slot circuit for the 14 LL144 intersectional categories.
- **Employer co-signed datasets**: Jubjub Schnorr attestation on the dataset hash so an auditor cannot substitute data.
- **On-chain disclosure receipts**: a regulator acknowledges an opening without revealing it.
- **Hosted proof server and DUST sponsorship**: auditors need no Docker and no tokens.
- **Lace-free verification embeds**: a one-line badge employers paste into job postings.

FairGate — the proof is public, the people are private.

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